

1.6 Rules of Inference - 2

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Note: A valid argument form is a tautology.

Consider Modus Tollens

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

The compound prop is false only when $T \rightarrow F$.

Then $\neg p = F$ or $p = T$.

$$\text{If } \neg q \wedge (p \rightarrow q) = T$$

both $\neg q = T$ and $p \rightarrow q = T$

so $q = F$. Then $p \rightarrow q \equiv T \rightarrow F$

which is F . So the argument form cannot be false i.e. is a tautology.

Rules of inference for quantified statements

Def: Universal instantiation is

concluding $P(c)$ is true given

$$\forall x P(x).$$

Choice of example.

"All frogs hop. Therefore Gregory the frog hops."

Def: Universal generalization is

making a universal conclusion $\forall x P(x)$

given $P(c)$ is true for every c .

Also true to say $P(c)$ is true if c is arbitrary.

Def: Existential instantiation making

the conclusion that c exists s.t. $P(c)$

given $\exists x P(x)$.

the conclusion that c exists s.t. $P(c)$
given $\exists x P(x)$.

Choice of example

"There is at least one frog that hops.

Therefore, there exists a frog that
hops, let's call it Gregory."

Def: Existential generalization is
concluding $\exists x P(x)$ given a particular
 c s.t. $P(c)$.

"Gregory the frog hops. Therefore, there
exists a frog that hops."

9) Universal Modus Ponens

$$(\forall x (P(x) \rightarrow Q(x)) \wedge P(a)) \rightarrow Q(a)$$

$P(x) = x$ is a frog

$Q(x) = x$ hops

$x = \{ \text{all frogs} \}$

$a = \text{Gregory the frog.}$

"All frogs hop. Gregory is a frog.
Therefore Gregory hops."

10) Universal Modus Tollens

$$(\forall x (P(x) \rightarrow Q(x)) \wedge \sim Q(a)) \rightarrow \sim P(a)$$

"A convertible is a fun car to
drive. Isaac's car is not a
convertible. Therefore Isaac's car
is not fun to drive."

$$(\forall x (P(x) \rightarrow Q(x)) \wedge \sim P(\text{Isaac's car})) \\ \rightarrow \sim Q(\text{Isaac's car})$$

To be valid this must be a tautology.

In order for this to be false, need

$T \rightarrow F$. So $\sim Q(c) = F$.

thesis

Fallacy of denying hypothesis

In order for this to be false, need

$T \rightarrow F$. So $\sim Q(c) = F$.

If $(\forall x (P(x) \rightarrow Q(x)) \wedge \sim P(c)) = T$

both $\forall x (P(x) \rightarrow Q(x)) = T$

and $\sim P(c) = T$. Then $P(c) = F$.

$\sim Q(c) = F \Rightarrow Q(c) = T$.

$P(c) \rightarrow Q(c) \equiv F \rightarrow T \equiv T$.

So the argument is false if there is

a c s.t. $P(c) = F$ & $Q(c) = T$.